SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1511

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**

Question Count: 4

Code of Conduct

I **NIVASINI VM 192511356** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Nivasini VM**

Assessment Tasks-1

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

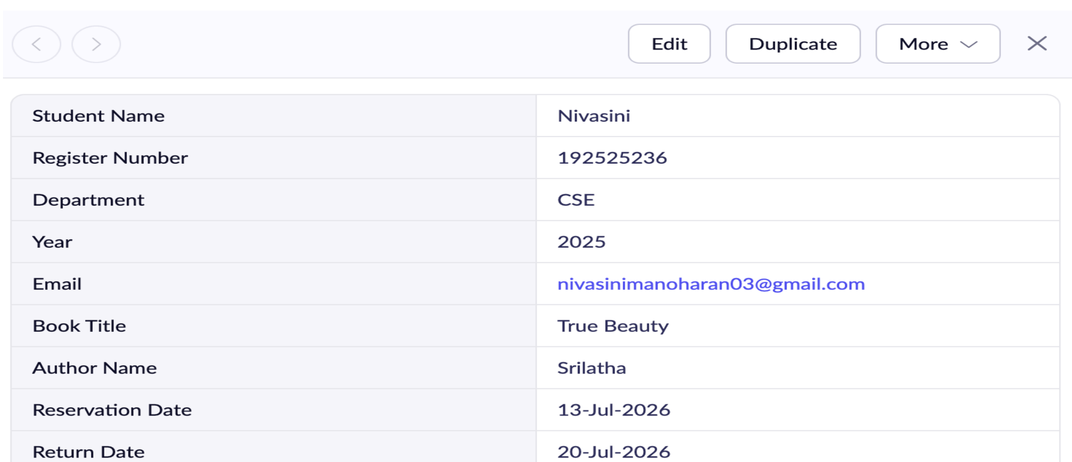
AIM

To create a simple cloud-based Library Book Reservation System for SIMATS Library using a Cloud Service Provider to demonstrate SaaS.

ALGORITHM

1. Start the application.
2. Enter Book Title, Author, Publication, Year, Edition, and Number of Copies.
3. Enter the Rack Number and book Status (Taken/Returned).
4. Store the book details in the cloud database.
5. Allow users to search and reserve available books.
6. Update the book status after reservation/return.
7. Display the updated book details.
8. Stop.

OUTPUT



Student Name	Nivasini
Register Number	192525236
Department	CSE
Year	2025
Email	nivasinimanoharan03@gmail.com
Book Title	True Beauty
Author Name	Srilatha
Reservation Date	13-Jul-2026
Return Date	20-Jul-2026

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM

To create a simple cloud-based Payroll Processing System using a Cloud Service Provider to demonstrate SaaS.

ALGORITHM

1. Start the application.
2. Enter Employee Name, Present Organization Experience, and Overall Experience.
3. Enter Basic Pay, HRA, DA, and TA.
4. Calculate Gross Pay.
5. Calculate Net Pay after applicable deductions.
6. Store the payroll details in the cloud.
7. Display the employee salary details.
8. Stop.

OUTPUT

Employee ID	34
Employee name	david
Department	HR
Present experience	3
Overall experience	5
Basic pay	₹ 30,000.00
HRA	₹ 6,000.00
TA	₹ 3,000.00
DA	₹ 2,000.00
Other_allowance	₹ 1,000.00
PF	₹ 2,000.00
Tax	₹ 1,000.00
Gross pay	₹ 42,000.00
Net pay	₹ 39,000.00

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

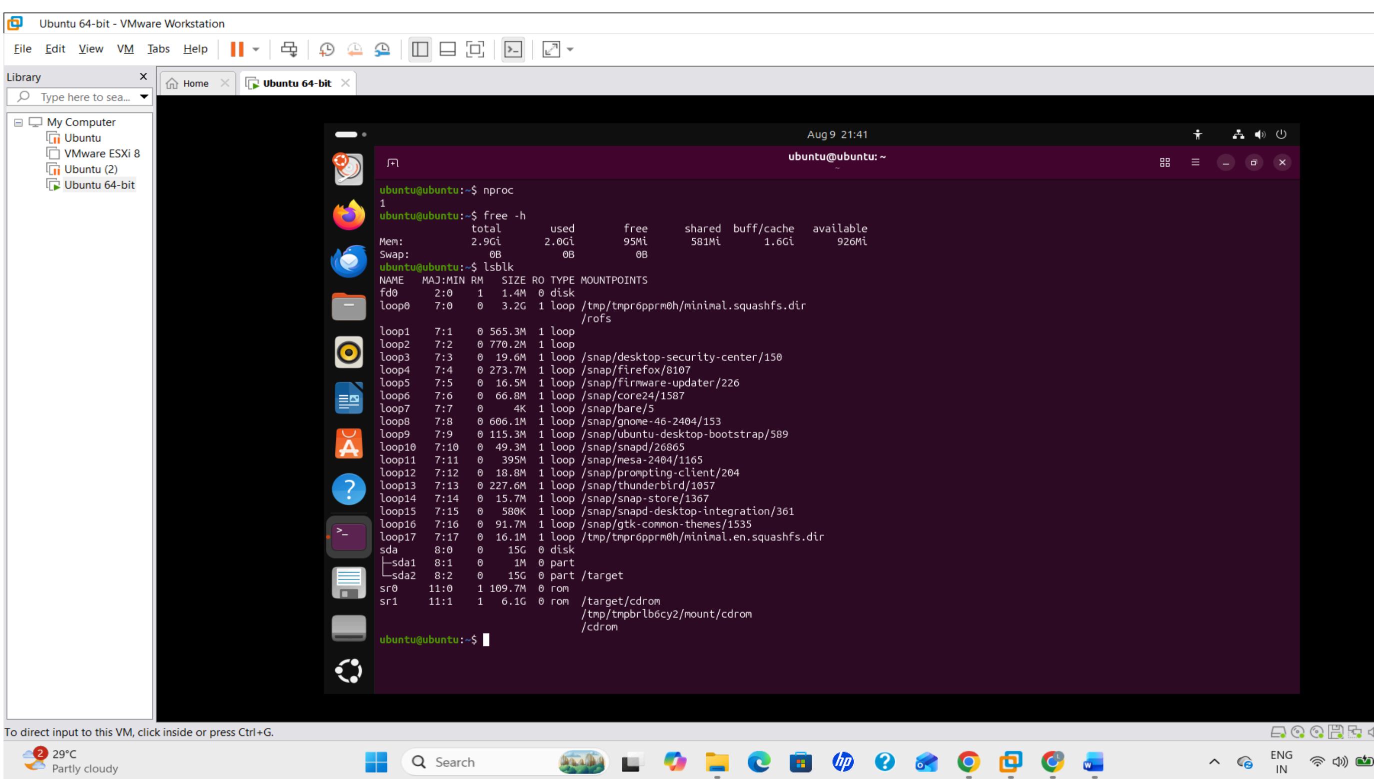
AIM

To demonstrate virtualization by installing a **Type-2 Hypervisor**, creating a virtual machine, and configuring it with **1 CPU, 2 GB RAM, and 15 GB storage**.

ALGORITHM

1. Install a **Type-2 Hypervisor** such as VMware or VirtualBox.
2. Create a new **Virtual Machine (VM)**.
3. Select the required **Windows/Linux ISO** as the guest OS.
4. Allocate **1 CPU, 2 GB RAM, and 15 GB storage**.
5. Install the guest operating system.
6. Start the VM and verify its configuration.
7. Stop the VM after successful execution.

OUTPUT



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

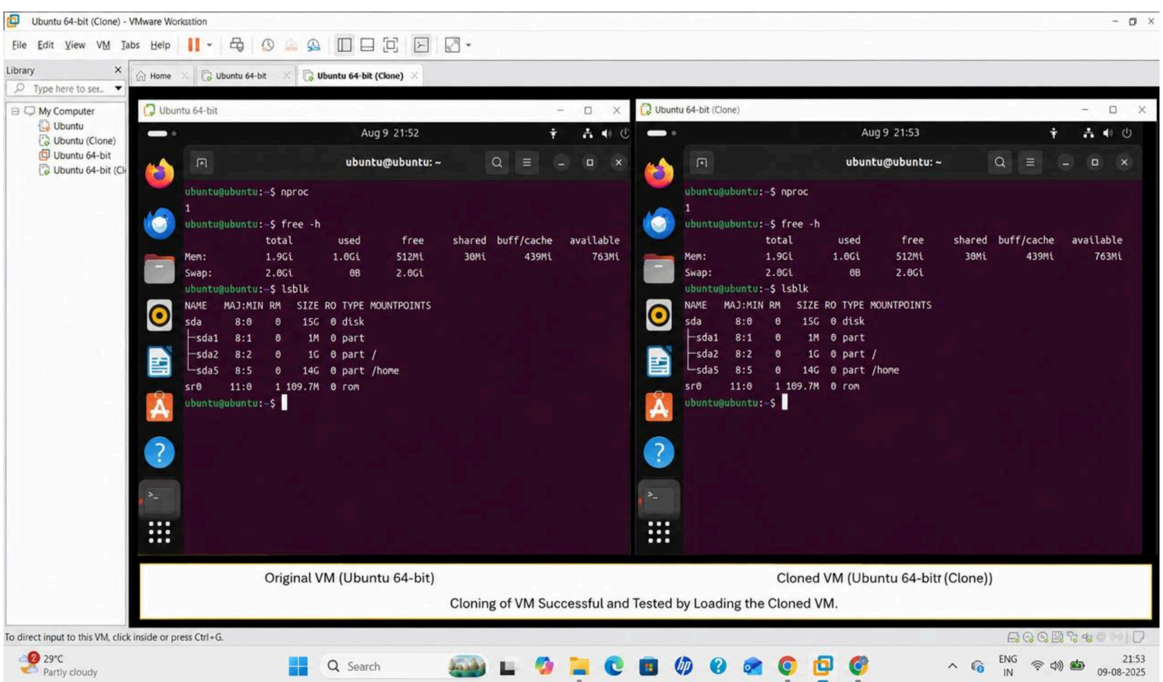
AIM

To demonstrate virtualization using a Type-2 Hypervisor by creating a virtual machine, cloning it, and testing the cloned VM/previous version.

ALGORITHM

1. Install and open a Type-2 Hypervisor such as VMware or VirtualBox.
2. Create and configure a Virtual Machine with Windows/Linux.
3. Install the required operating system and applications.
4. Shut down the VM and create a clone of the VM.
5. Assign a new name to the cloned VM.
6. Start the cloned VM and verify its operating system and files.
7. Compare the original and cloned VM to verify successful cloning.
8. Stop the VMs after testing.

OUTPUT



Assessment Tasks-2

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

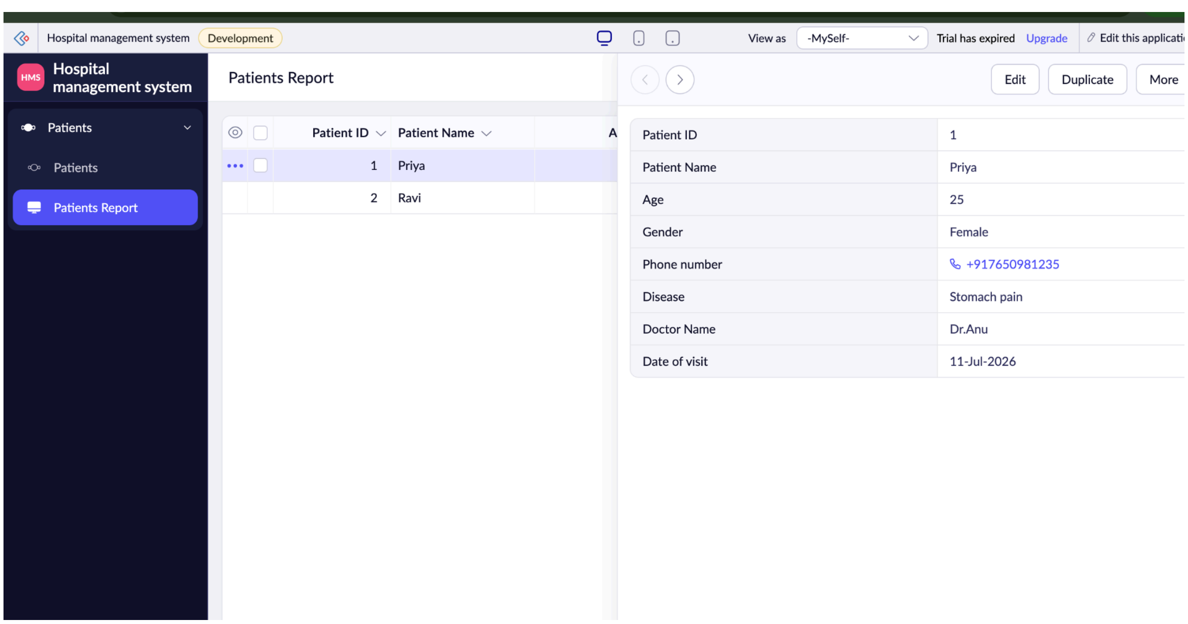
AIM

To create a simple **cloud-based Hospital Information System** for SIMATS Hospital using a Cloud Service Provider to demonstrate **SaaS**.

ALGORITHM

1. Start the application.
2. Enter Patient Name, Age, Address, Phone, and Email.
3. Enter Attender Name and Attender Phone.
4. Enter Doctor Name and Disease details.
5. Store the patient information in the cloud database.
6. Display and manage the patient details.
7. Stop the application.

OUTPUT



Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

AIM

To create a simple **cloud-based Online Supermarket application** using a Cloud Service Provider to demonstrate **SaaS**.

ALGORITHM

1. Start the application.
2. Enter **Customer Name, Address, Phone, and Email**.
3. Select **Item Classification**.
4. Enter **Item Name and Quantity**.
5. Store the customer and order details in the cloud.
6. Display the selected items and order details.
7. Stop the application.

OUTPUT

Customer name	NIVASINI
Address	ANNA NAGAR ,CHENNAI
Phone	+919876543210
Email	niva@gmail.com
Item	Vegetables
Classification	Grocery
Quantity	2

Add a comment

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

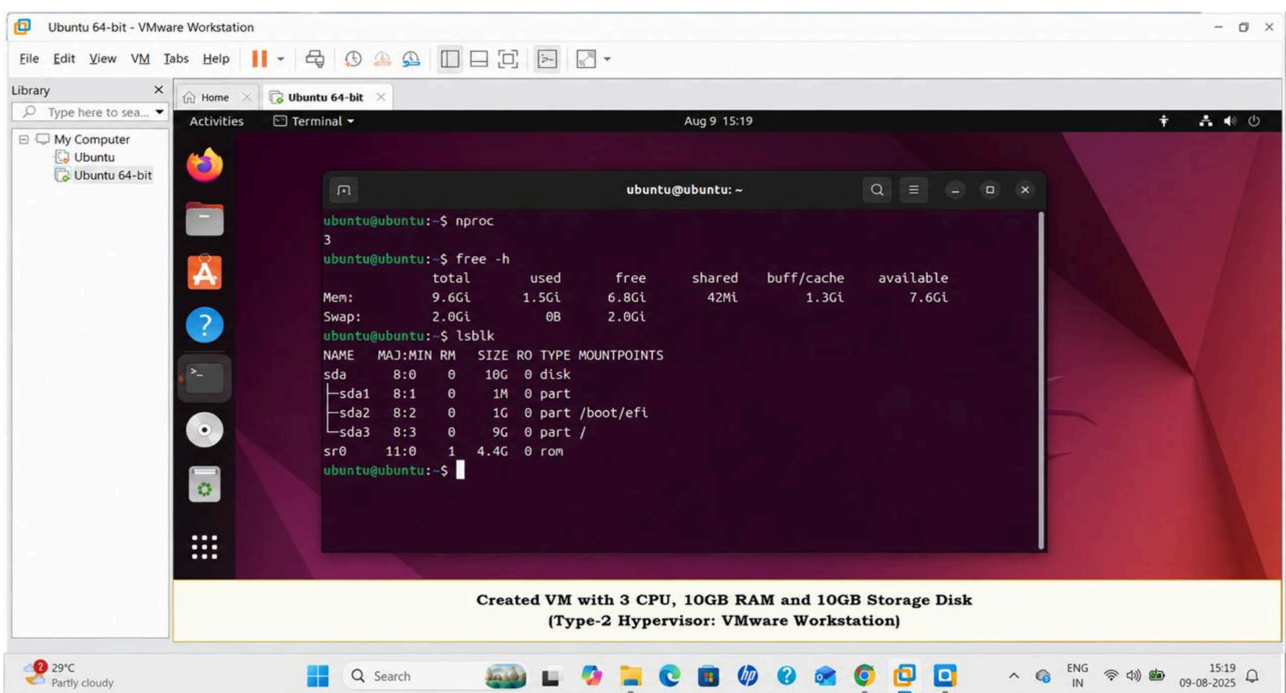
AIM

To demonstrate virtualization by installing a **Type-2 Hypervisor**, creating a virtual machine, and configuring it with **3 CPUs, 10 GB RAM, and 10 GB storage**.

ALGORITHM

1. Install and open a **Type-2 Hypervisor** such as VMware or VirtualBox.
2. Create a new **Virtual Machine (VM)**.
3. Select the required **Windows/Linux ISO**.
4. Allocate **3 CPUs, 10 GB RAM, and 10 GB storage**.
5. Install the guest operating system.
6. Start the VM and verify the configuration.
7. Stop the VM after successful execution.

OUTPUT



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

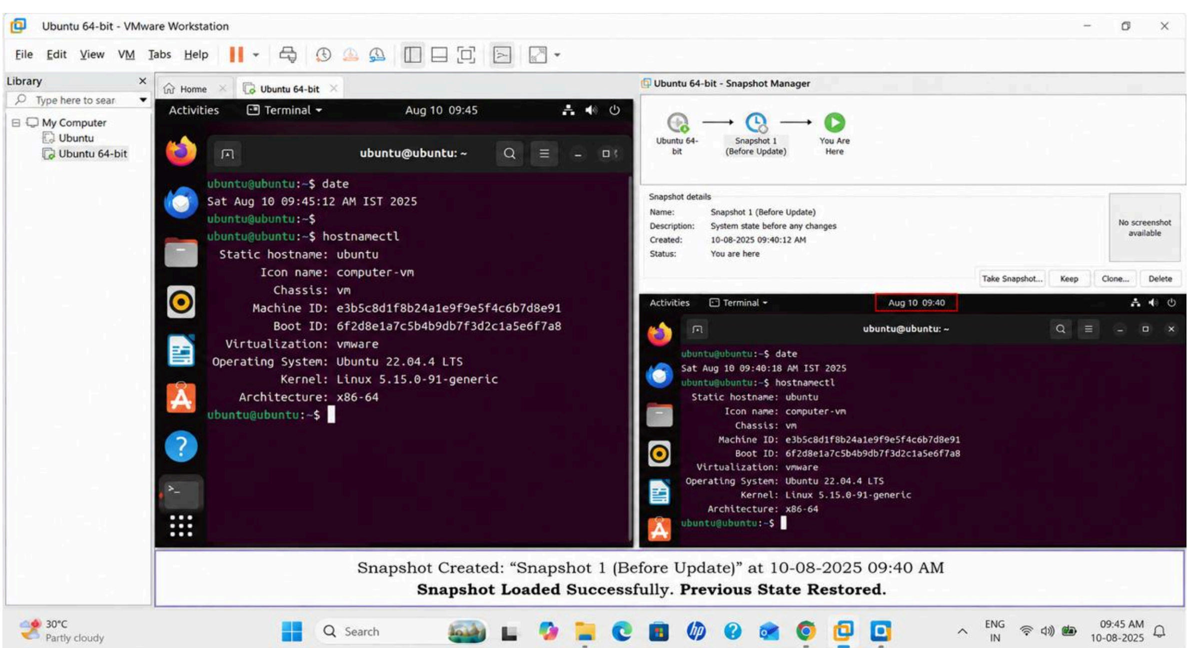
AIM

To demonstrate virtualization using a **Type-2 Hypervisor** by creating a VM, taking a **snapshot**, and testing it by restoring the previous version.

ALGORITHM

1. Install and open a **Type-2 Hypervisor** such as VMware or VirtualBox.
2. Create and configure a **Windows/Linux Virtual Machine**.
3. Install the required operating system and applications.
4. Take a **snapshot** of the VM.
5. Make some changes to the VM.
6. Restore the VM using the saved snapshot.
7. Verify that the VM has returned to its **previous version/state**.
8. Stop the VM.

OUTPUT



Assessment Tasks-3

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

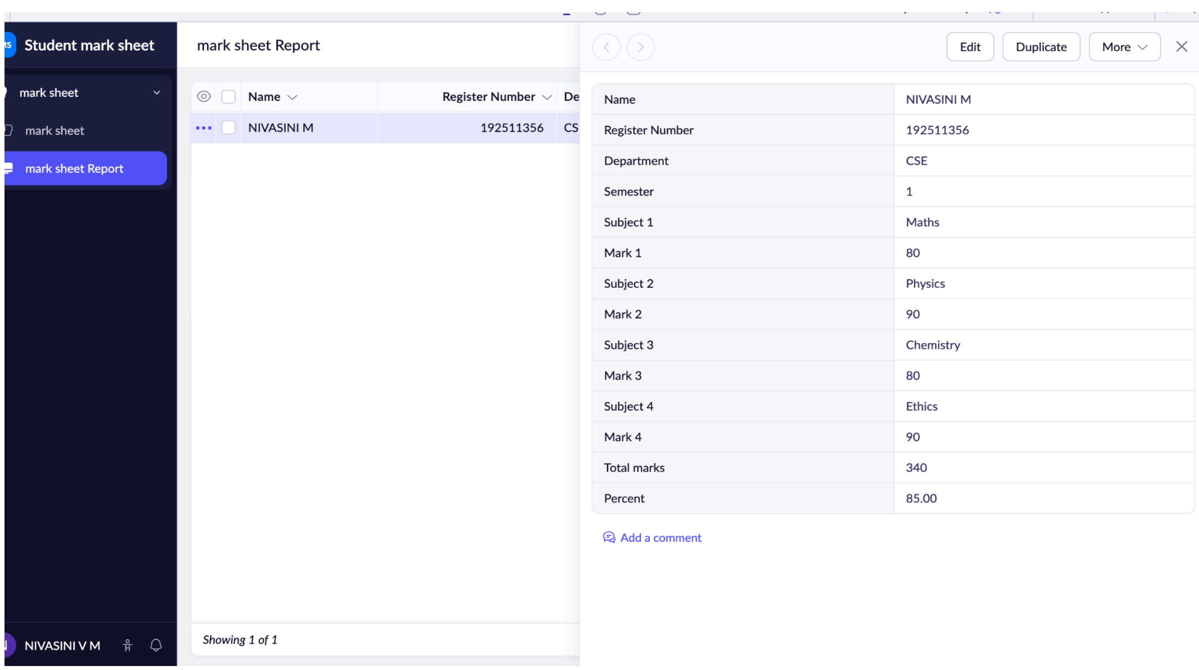
AIM

To create a simple cloud-based Mark Sheet Processing System using a Cloud Service Provider to demonstrate SaaS, including student details, marks, average, percentage, rank, and overall performance.

ALGORITHM

1. Start the application.
2. Enter student Name, Reg No., Subjects, and Marks.
3. Calculate Total, Average, and Percentage.
4. Calculate the Rank based on marks.
5. Determine Overall Performance.
6. Store and display the mark sheet through the cloud application.
7. Stop.

OUTPUT



Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

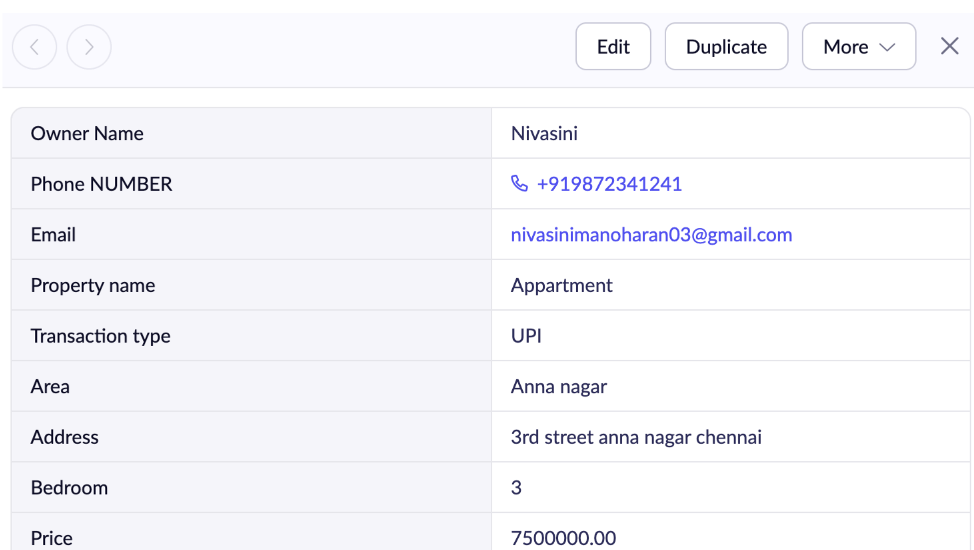
AIM

To create a simple **cloud-based Property Buying and Rental System** for properties in Chennai using a Cloud Service Provider to demonstrate **SaaS**.

ALGORITHM

1. Start the application.
2. Enter **Property Type (Buy/Rent/Commercial)**.
3. Enter **Area, Place, Range, and Property Value**.
4. Enter **Rental Agreement** details for rental properties.
5. Store the property details in the cloud database.
6. Allow users to search and select suitable properties.
7. Display the selected property details.
8. Stop.

OUTPUT



Owner Name	Nivasini
Phone NUMBER	+919872341241
Email	nivasinimanoharan03@gmail.com
Property name	Appartment
Transaction type	UPI
Area	Anna nagar
Address	3rd street anna nagar chennai
Bedroom	3
Price	7500000.00

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

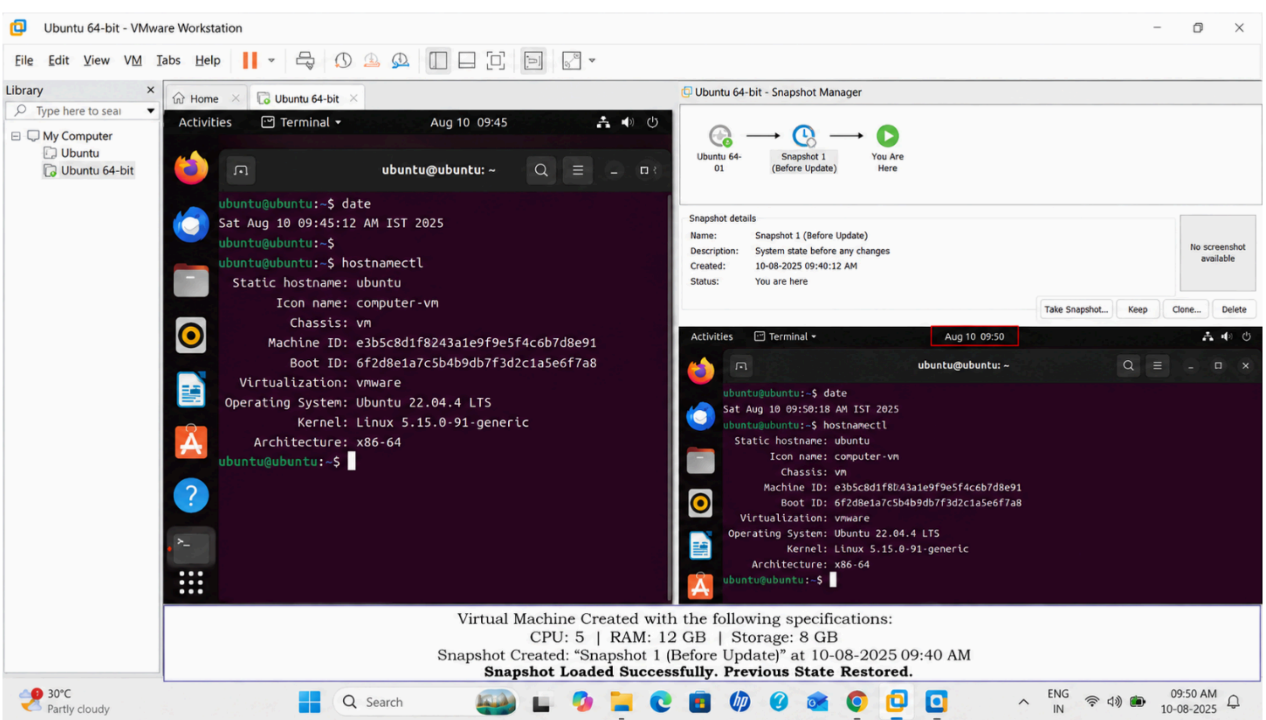
AIM

To demonstrate virtualization by installing a **Type-2 Hypervisor** and creating a virtual machine with **5 CPUs, 12 GB RAM, and 8 GB storage**.

ALGORITHM

1. Install and open a **Type-2 Hypervisor** such as VMware or VirtualBox.
2. Create a new **Virtual Machine (VM)**.
3. Select the required **Windows/Linux ISO**.
4. Allocate **5 CPUs, 12 GB RAM, and 8 GB storage**.
5. Install the guest operating system.
6. Start the VM and verify the configured resources.
7. Stop the VM after successful execution.

OUTPUT



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

AIM

To demonstrate virtualization using a **Type-2 Hypervisor** by creating a VM, taking a **snapshot**, and testing it by restoring the previous version.

ALGORITHM

1. Install and open a **Type-2 Hypervisor** such as VMware or VirtualBox.
2. Create and configure a **Windows/Linux Virtual Machine**.
3. Install the required operating system.
4. Take a **snapshot** of the VM.
5. Make changes to the VM.
6. Restore the VM using the saved snapshot.
7. Verify that the **previous version/state** is restored.
8. Stop the VM.

OUTPUT